Laboratory Prototype Scorecard Spreadsheet Review

Jason Glazer, GARD Analytics, Inc.

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Summary of Major Comments

The following comments represent errors found when comparing values from the scorecard to the source cited in the scorecard:

- Wrong roof requirement in CZ10 (LAB27-Envelope)
- SHGC for 1998-2007 is wrong value (LAB28-Envelope)
- Parking area lighting power calculated wrong (LAB66-Ext Lights)
- One weekday office occupancy schedule value is incorrect (LAB82-Schedules)

The following are the highest priority comments:

- What specific type of lab is the prototype (LAB4-Notes, LAB43-Ventilation)?
- The building area is many times larger than typical (LAB11-Prototype).
- A ribbon of glass on each floor representing windows is unlikely (LAB14-Prototype).
- Lab buildings have types of spaces besides the labs and offices (LAB19-Zones).
- Three air handling units for this size lab building is too small (LAB21-Zones).
- The occupant density for the lab spaces is too high (LAB22-Zones).
- Total static pressure seems low (LAB34-HVAC System).
- Calculated exhaust flows to low (LAB41-Ventilation, LAB42-Ventilation).
- Other types of exhaust besides fume hoods are missing (LAB44-Ventilation).
- Use of additional lighting power is likely in a lab (LAB59-Interior Lights).
- How are the fume hood sash position schedules intended to be used (LAB46-Ventilation and LAB85-Schedules)?
- Lab hot water and elevator schedules should not be zero (LAB87-Schedules).

The following comments represent other important issues found:

- A wide range of sources are available and could have been used (LAB1-Notes)
- The volume above the ceiling may not be large enough for a lab (LAB12-Prototype)
- The exhaust system configuration is not described (LAB36-HVAC System).
- The exhaust hood flows do not impact the ventilation (LAB39-Ventilation).
- The fume hood flows should not be based on kitchen hoods (LAB40-Ventilation).

- The minimum required exhaust rate is usually much lower than the exhaust rate in a lab (LAB45-Ventilation).
- A bottom-up analysis of ventilation requirements based on specific equipment should be used to estimate system flows (LAB47-Ventilation)
- Exterior lighting power seems incorrect (LAB63, LAB65, LAB67-Ext Lights).
- Lab plug loads should be determined from bottom-up analysis (LAB-81-Equipment).

In addition, the following comments describe the lack of disclosed assumptions that make verifying the values in the scorecard nearly impossible:

- Why were the vintages chosen the way they were (LAB8-Notes)?
- What is the original source of geometric assumptions (LAB9-Prototype)?
- No justification is provided for the area and other major assumptions (LAB7-Notes)
- How was the number of floors chosen (LAB13-Prototype)?
- How was the fraction of support and office spaces determined (LAB20-Zones)?
- How was the zoning of labs and offices determined (LAB18-Zones)?
- How were the wall and roof type chosen (LAB29-Envelope)?
- How was System 14a selected (LAB32-HVAC System)?
- How was the new construction fan power determined (LAB38-HVAC System)?
- Why no setback/setup in lab spaces (LAB48-Thermostat)?
- What are the space humidity requirements (LAB50-Thermostat)?
- Where does the refrigeration power density come from (LAB73-Equipment)?
- Where does the elevator power come from (LAB78-Equipment)?

Background

The review of the "score card," which is a spreadsheet summarizing the model assumptions, is broken down by spreadsheet tab, which includes:

- Notes
- Prototype
- Zones
- Envelope
- HVAC System
- Thermostat
- Ventilation
- Water Heater
- Interior Lighting

- Ext Lighting
- Equipment
- Schedules
- PV
- EV Charger

The comments on these tabs are labeled using the following format **LABx**-<TAB>-<CELLRANGE>), where <TAB> is the name of the tab in the spreadsheet and <CELLRANGE> indicates the specific cells. Note that no tabs were reviewed after the Schedules tab.

A copy of the spreadsheet is available that includes which cells on each tab were checked against the sources and were consistent (shown in light green) and which had anomalies and are described in a comment below (shown in tan).

LAB1-Notes (n/a)

To provide a thorough review of the prototype, I reviewed several great resources shown below prior to beginning my review. Originally, 70 resources were identified, but the following seemed the most important.

Sources that *should* be reviewed for background information and are fundamental for reviewing the laboratory prototype are the following:

- *ASHRAE Handbook HVAC Applications 2023 Chapter 17 "Laboratories"* Essential for understanding laboratory HVAC and covers almost many important topics, although briefly. If only one thing is reviewed, this should be it. https://store.accuristech.com/standards/2023-ashrae-handbook-hvac-applications-i-p?product_id=2225673
- Market Assessment of Energy Efficiency Opportunities in Laboratories. Alison Paradise 2015. Contain statistics on lab sizes in California, laboratory equipment included, and estimated plug loads. Average size of labs by subtype. https://www.etcc-ca.com/sites/default/files/reports/ceel_market_assessment_et14pge7591.pdf
- Boston Area Laboratory Energy Benchmarking Study. 2017. Survey of a large number of labs in Boston that might provide some insight into overall energy use. Shows statistics for the number of different laboratory types. https://greenribboncommission.org/wp-content/uploads/2017/09/GRC-HEWG-Labs-Benchmarking-Analysis-Year-2_REV.pdf

- High Performance Laboratories – A Design Guidelines Sourcebook. 2011. PG&E. Contains standard, good, and better values for ventilation, other HVAC, and some modeling parameters https://www.pge.com/assets/pge/docs/save-energy-and-money/rebate-and-incentives/Labs_BestPractices.pdf
- Laboratory Benchmarking Tool – a website to provide benchmarking and includes EUIs, peak load, and demand for overall, ventilation, lighting, plug loads, and water for over 1400 labs. Similar to Portfolio Manager. https://lbt.i2sl.org/buildings/charts
- Prototype Laboratory Concept. Sandia National Labs. 2015. Space programs. Many specific layouts for different "modules" of a lab building. List of typical spaces. https://www.osti.gov/biblio/1227344
- High Efficiency Fume Hoods in Laboratory Spaces – Results Report. CASE. 2018. Contains statistics and modeling assumptions. Already listed in the scorecard. Prototype model summary. Lab area by type. Average sash position. https://title24stakeholders.com/wp-content/uploads/2019/01/T24-2019-CASE-Study-Results-Report-Fume-Hoods_Final_with_Attachments.pdf
- Characterizing the Laboratory Market. LNBL 2017. Includes the average size of labs. https://www.osti.gov/servlets/purl/1350977

Additional useful sources could be added to the list shown in the scorecard spreadsheet:

- Raising the Bar in High Performance Laboratory Design, Jeffery M. Scot – Full article has lab layouts. https://www.ashrae.org/technical-resources/ashrae-journal/featured-articles/raising-the-bar-in-high-performance-laboratory-design
- Laboratories for the 21st Century: Energy Analysis. 2003. Describes a model and its assumptions, including schedules, lighting, and equipment power density, uses DOE-2.2, details on mechanical and simulation parameters, and mechanical system details. https://docs.nrel.gov/docs/fy03osti/33410.pdf
- ASHRAE Laboratory Design Guide - 2nd Edition. More detail than the handbook chapter, but not many specifics for modeling. https://store.accuristech.com/standards/ashrae-laboratory-design-guide-planning-and-operation-of-laboratory-hvac-systems-2nd-ed?product_id=1897560
- Building Type Basics for Research Laboratories Second Edition. From 2008. It contains a good description of programming and types of spaces. Page 72 has the typical time per day for researchers for different activities.

- Classification of laboratory ventilation design levels. 2018. ASHRAE. Shows what HVAC options are acceptable for different classifications from 0 to 4. https://www.ashrae.org/File%20Library/Technical%20Resources/Free%20Resources/Publications/ClassificationOfLabVentDesLevels.pdf
- Design Requirements Manual. National Institutes of Health. 2016. 1200 pages of laboratory specifications. Includes room data sheets. https://orf.od.nih.gov/TechnicalResources/Documents/DRM/DRM1.503262020.pdf
- Lessons Learned from Laboratory-Specific Optimization Technologies for an Existing Campus and a New Facility with a Conserved Heritage Façade. Larissa Ide. Contains details from modeling and a good end-use breakdown. https://publications.ibpsa.org/proceedings/esim/2022/papers/esim2022_228.pdf
- Guidelines for Laboratory Design. 2013. This book has sections on each type of laboratory and on HVAC. Matrix of building considerations for each type of lab. https://onlinelibrary.wiley.com/doi/book/10.1002/9781118633816
- I2SL – International Institute for Sustainable Laboratories https://www.i2sl.org/best-practices
  Contains hundreds of pages of best practices on the website, including:
  - An Introduction to Low-Energy Laboratory Design
  - Benchmarking Energy Efficiency in Laboratories
  - Laboratory Guidelines Using ASHRAE 90.1—2019 Appendix G
  - Evaluating Laboratory Design: National Institutes of Health Building 50
- Does the Lab Building Follow the Model? Kelley Cramm. ASHRAE Journal 2021. Contains occupancy schedules from a specific lab. https://technologyportal.ashrae.org/Journal/ArticleDetail/2316
- HVAC Resources Map for Laboratories – links to many other resources https://hvacresourcemap.nlr.gov/laboratories
- Self-benchmarking Guide for Laboratory Buildings: Metrics, Benchmarks, Actions. LBNL-3391E. Metrics for whole building, ventilation, cooling, heating, plug, and lighting. 2009. https://eta-publications.lbl.gov/sites/default/files/lbnl-3391e.pdf
- Smart Lab Toolkit Resources: https://smartlabs.i2sl.org/resources.html
- Laboratory Systems: Impact of Functional, Operational, and Climatic Parameters. 2014. Modeling typical values and schedules. https://publications.ibpsa.org/proceedings/simbuild/2014/papers/simbuild2014_20.pdf

LAB2-Notes (n/a)

Rather than repeat it, please see comment PPD1 in my single-family residential prototype review, LHSC1-Notes in the large hotel review, and SHSC1-Notes in the small hotel review.

LAB3-Notes (n/a)

Rather than repeat it, please see comment PPD2 in my single-family residential prototype review, LHSC2-Notes in the large hotel review, and SHSC2-Notes in the small hotel review.

LAB4-Notes (n/a)

Laboratories come in many different types with very different ventilation and internal loading, and there is no indication of which type this prototype in the scorecard is intended to represent. Without this information, it is difficult to evaluate the reasonableness of many of the assumptions documented in the scorecard. The types of laboratories include:

- Biological
- Chemical
- Animal
- Physical

In addition, ASHRAE has classifications for Laboratory Ventilation Design Levels, numbered 0 to 4, that correspond from negligible to extreme levels of hazard severity. Biosafety labs have a similar classification scheme from 2 to 4.

Other ways to organize different types of labs include:

- Academic teaching laboratories
- Academic research laboratories
- Life science research company laboratories
- Hospital research facility laboratories

Some subtypes of laboratories from the ASHRAE Laboratory Design Guide are:

- Biological laboratories
    - Animal laboratories
    - Biological containment laboratories using infectious agents
    - Hospital laboratories
- Chemical laboratories
    - Synthesis laboratories

    - Analytical laboratories
    - Radiochemistry laboratories
    - Teaching laboratories
- Physical laboratories
    - Cleanrooms
    - Materials testing laboratories
    - Nanotechnology/electronics/instrumentation laboratories

If the intention of the prototype is to represent all laboratories, it would be appropriate to disclose the weighting percentage used for each type and the assumptions used for each type so that the weighted value is seen as appropriate. Given that multiple zones are probably appropriate for a laboratory prototype, perhaps each zone could represent each of these specific types of laboratories. They would approximate the laboratory market better and would better show the impact of various measures that may be applied in the future that may be appropriate for some types of labs.

LAB5-Notes (B2:B4, B9:B12)

A number of data sources do not show a URL or a more detailed source in order to find it, including:

- Dodge 2023
- CEUS 2022
- ComStock 2023
- Title 24, part 6, referred to as T24 (1984, 2001, 2007, 2013, 2016, 2022, 2025)
- ASHRAE 90.1 (2022)
- CFR Title 10 (2008, 2010, 2025)
- ACM 2022

LAB6-Notes (B3:B4)

It is not clear that the references below are ever used:

- CEUS 2022
- ComStock 2023

Please confirm that all the references cited are used and remove any that are missing.

LAB7-Notes (n/a)

No justification is provided for the overall size of the facility that has been included with other scorecards. The area of the floorplate, the number of stories, the window-to-wall ratio, floor-to-floor height, and other major parameters are not justified.

LAB8-Notes (B21:D32)

The categorization of the different ages of construction for the various vintages is not provided, even though a graph of construction rates is shown. That graph shows significant lab construction in the 1980s and early 1990s, with another bump of activity in the late 1990s and early 2000s, and some more recent activity in the early 2020s.
These trends for other building scorecards have been translated into various vintage assumptions.

LAB9-Prototype (A7:C16)

Almost all the assumptions on this tab are based on the CEC (presumably the CEC Laboratory prototype from CBECC 2022.3.1 software referenced from the Notes tab), but none of these assumptions seem to have been further justified. What is the basis of the CEC lab prototype assumptions? Are they still valid today?

The assumptions citing CEC include:

- Shape
- Total Floor Area (ft^2)
- Aspect Ratio
- Number of Floors
- Roof type
- WWR
- Shading Geometry
- Floor to floor height (feet)
- Floor to ceiling height (feet)
- Glazing sill height (feet)

Greater justification should be provided and not simply accepted because a previous prototype used them.

LAB10-Prototype (B31)

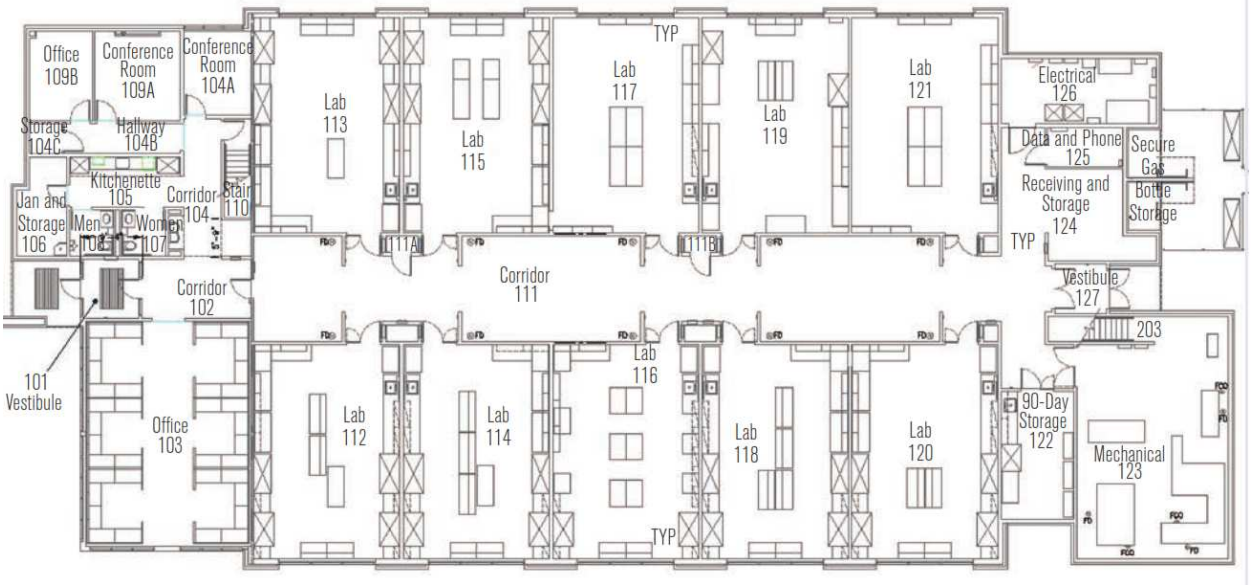
The DEER Manufacturing-BioTech model is one of two existing prototypes cited on the scorecard, and yet no assumptions are provided for this prototype. In fact, the area of the prototype is not even shown. It would be better to list the major assumptions for the two prototypes as well as other sources (see LAB1-Notes for other possible sources). By comparing across sources and recommending a specific value for an assumption, it is much easier to accept the validity of that assumption.

LAB11-Prototype (B8)

The total floor area of the prototype is 53,633 sqft, which is only justified based on "Same as CEC," and this seems much larger than the values found looking at a few of the sources (see bullets below). A floor size between 4,000 and 15,000 sqft seems more justifiable.

- The PNNL lab, which is 14,700 sqft, is a single-story wet chemistry lab from "Raising the Bar in High Performance Laboratory Design." Jeffery Scott. ASRHAE Journal June 2020.

FIGURE 1 Floor plan of the 14,700 ft² (1366 m²) wet chemistry research facility.



- From the Market Assessment of Energy Efficiency Opportunities in Laboratories, the results are based on a survey done with only 23% of the respondents able to estimate the area of the lab. Note that LSR is “life science research” in the table below. It was assumed that the responses did not include the laboratory support

spaces, so after adjusting for support spaces:

TABLE 74: ESTIMATED NUMBER OF LABORATORIES (ADJUSTED) BY MARKET SEGMENT IN CALIFORNIA

MARKET SEGMENT	ESTIMATED AVERAGE SQ FT (M) PER LAB	ESTIMATED NUMBER OF LABS
Academia	4,000	6,200
LSR	5,800	11,700
Hospitals	4,600	1,700
Other	5,700	540
Total		**20,100**

- Characterizing the Laboratory Market. The average lab size in the U.S. is 4,899 sqft. By category, that is life science research average of 5809 sqft, research institutions at 4297 sqft, teaching labs at 4304 sqft, and clinical diagnostic labs at 3580 sqft. This study is focused on life science labs.

Table 4: Floor space and laboratory counts for laboratories associated with life science research, research institutions, teaching, and clinical diagnostic.

Market Segment	Floor space (millions)	Numbers of labs
Life Science Research	359	61,800
Research Institutions	223	51,900
Teaching	164	38,100
Clinical Diagnostic	5.37	1,500
Total	**751**	**153,300**

LAB12-Prototype (B14:B15)

The floor-to-floor height of 13 feet and the floor-to-ceiling height of 9 feet do not seem sufficient for portions of the exhaust system and HVAC system that would be located above the ceiling. The ceiling above lab spaces is often very crowded with ductwork and especially in multi-floor lab may have large horizontal as well as vertical ducts.

LAB13-Prototype (B10)

Due to the complexity of the exhaust systems and specifically fume hoods, one common laboratory configuration is a single story, but in large academic and industrial laboratories, multiple floors are common. It would be good to look through sources, including those

provided in LAB1-Notes, and determine if the three floors assumed in the scorecard are reasonable.

LAB14-Prototype (B12)

The uniform 33% window-to-wall ratio on all orientations of the laboratory building is very unlikely. It is also unlikely that the windows would appear as long ribbons of glass on each face. The laboratory rooms themselves are likely to have punched openings and may have good daylighting, as would offices, but other support rooms, such as for storage, mechanical, electrical, and receiving, would likely have fewer windows. Typically, individual laboratories are long rooms off a central hallway with windows on one end and work tables along each side, which often include fume hoods. Fume hoods are very unlikely to be directly adjacent to windows. Generally, punched windows are probably more inexpensive to build than curtainwalls that are implied by the uniformity of the ribbon of glass. The window-to-wall ratio should be sensitive to the space type of the room that is adjacent to it. Also, the 33% may be simply leftover from a prototype for an office building that had that same percentage. If that is the case, the percentage should be re-examined for the laboratory building based on the sources.

LAB15-Prototype (n/a)

It would be good to disclose the overall parameters of the laboratory building, such as the number of workbenches and the number of fume hoods. Also, the fraction of the building being used directly as lab spaces versus support spaces and office spaces should be disclosed.

LAB16-Prototype (A28)

It would be great if multiple prototypes were being compared here, like other scorecards, but instead, the DEER model seems to be getting ignored. Please summarize the DEER model and the CEC Lab model, as has been done on other scorecards, and let the assumptions that are most justifiable be used.

LAB17-Zones (L2:O21)

The diagram of the building is shown as a simple rectangle, but I believe this is a building model that uses a rectangular core zone and trapezoidal perimeter zones. Please show the actual layout in the diagram.

LAB18-Zones (L2:O21)

Based on the previous comment, the use of such simplified core and perimeter zones is not appropriate for a prototypical model that should instead derive the areas for all the spaces based on statistics and knowledge and then combine the spaces in a layout that is reasonable and consistent with common practice for the building type.

LAB19-Zones (n/a)

The variety of spaces shown is not representative of a laboratory. The prototype includes two space types:

- Office Area (>250 square feet)
- Science laboratories

An actual laboratory would include many more spaces, such as:

- Secure entry
- Office
- Lab entry
- Washroom
- Breakroom
- Multi-purpose room
- Diagnostics lab
- Sample receipt
- Clinical collection
- Molecular suite
- Histology lab
- Postmortem
- Biochemistry
- Clean media lab
- Cleaning and sterilization
- Equipment lab
- Open lab

- Sample storage
- Loading and storage
- Electrical
- Plumbing
- Effluent decontamination
- Corridor
- Enclosed mechanical

This is based on the "Prototype Laboratory Concept" by Sandia National Laboratory. While some of these could be aggregated into a general concept of "science laboratories," most of these spaces should be separately zoned. This list does not include typical spaces for animal-related labs or in academic teaching labs.

LAB20-Zones (E3:20)

27% of the laboratory building is offices, and 73% is "science laboratories." These fractions are very important since lab spaces will have a much higher internal loading and ventilation requirements. This fraction has not been justified, and given how important it is to the energy consumption of the building, it should come from a citable source.

In the Prototype Laboratory Concept document by Sandia National Labs, they provided an example building to demonstrate functional space programming. In that example, approximately 58% of the area was lab space, 18% was admin, and 23% were mechanical spaces and corridors:

Functional Space Program

Spaces Required		Quantity	Area per Space M^2	Total area M^2
Administrative and Amenity Spaces				
A.01A	Secure Entry Module A	1	25	25
A.02D	Office Module D	1	45	45
A.03B	Lab Entry Module B (double with showers)	1	30	30
A.05A	Break Room Module A	1	25	25
		Subtotal Amenity Areas		**125**

Spaces Required		Quantity	Area per Space M^2	Total area M^2
Laboratory Spaces				
L.01B	Diagnostics Lab Module B (single 1 BSC)	3	25	75
L.01F	Diagnostics Lab Module F (double, bench only)	1	50	50
L.02A	Sample Receipt Module A	1	25	25
L.04C	Molecular Suite Module B	1	50	50
L.06C	Post Mortem Module C	1	25	25
L.09B	Cleaning & Sterilization Module B	1	25	25
L.10A	Equipment Lab Module A (bench & floor equip)	1	25	25
L.11B	Open Lab Module B	2	50	100
L.12A	Sample Storage Lab Module	1	25	25
		Subtotal Lab Areas		**400**

Spaces Required		Quantity	Area per Space M^2	Total area M^2
Support Spaces				
S.01C	Loading & Storage Module C	1	75	75
S.02A	Electrical Room Module A	1	15	15
S.03A	Plumbing Room Module A	1	15	15
S.05	Corridor (1.8m wide)	19	3.06	58
		Subtotal Support Areas		**163**
		Total Interior Area		**688**

LAB21-Zones (C3:C20)

Based on the descriptions, the building has only three air handling units, each serving every space on a specific floor. The top and bottom floors are a mixture of laboratory and office spaces. For those two floors, it is more likely that separate air handling units would be used for the laboratories and offices. The complex ventilation and pressurization requirements of laboratories and sometimes different hours of operation would usually result in separate air handling units serving those spaces.

LAB22-Zones (I3:J5, I12:J15)

The occupant densities shown in column J are consistent with Table 120.1-A of Title 24-2025 as cited. For laboratory spaces and office spaces, these are based on the occupancy rate assumed for ventilation. For labs, this is 25 people/1000 sqft or 40 sqft/person. To make sure that adequate ventilation is designed, these numbers are sometimes very conservative. For laboratories, these are probably not appropriate since the ventilation requirements driven by the design exhaust for fume hoods, snorkels, or other laboratory exhaust equipment may be driving the need. Other sources for determining the occupant density:

- Count the number of fume hoods since the typical occupancy of a laboratory is driven primarily by workstations, including fume hoods.
- For LEED 4.1, the recommended occupant density is 400 sqft/person (see https://www.usgbc.org/credits/new-construction-existing-buildings-commercial-interiors-core-and-shell-schools-new-constr-3)
- From Final Measure Information Template – Laboratory Exhaust VAV and Reheat CASE report: 100-200 sqft/person
- From Laboratories for the 21st Century Energy Analysis: 275 sqft/person

Additional references from the list of sources in LAB1-Notes may be found, but in general, it seems likely that 40 sqft/person is unrealistically dense.

The overestimate for the number of occupants is furthered by the lack of corridors, restrooms, and other low-occupant-density rooms such as mechanical and electrical rooms.

Also note, T24, 2022, Appendix 5.4A, sheet: "2022 SpaceFuncData-Input" that is used on the water heater, equipment, and schedules tab uses 10 people/1000 sqft for "Laboratory, Scientific" category, not 25 people/1000 sqft.

LAB23-Zones (n/a)

Pressurization relationships between zones in laboratory buildings are very important to the design and layout of spaces and should be understood and disclosed.

LAB24-Zones (n/a)

For the prototype, 73 people work in office support roles compared to the 976 who work in laboratory spaces. This is approximately 13 lab workers for each office worker. This number seems too large and would expect more office workers to support laboratory work. This is

especially true in clinical labs that have a high rate of samples flowing into the lab for testing.

LAB25-Zones (n/a)

Please add a table that summarizes the total area and fraction of area by the space type, as well as the total number of people and fraction of people by space type.

LAB26-Zones (n/a)

Please add the type of labs for each lab zone and the number of fume hoods or other exhaust equipment for each.

LAB27-Envelope (R8:R9)

The roof u-value requirements for the "pre 1978" and "1978-1997" vintages are shown generally as the same as the values in the "1998-2007" vintage, with the note "Considering 30 years for roof/ insulation replacement." This is consistent except for CZ10, which showed 0.057 for the "1998-2007" vintage value and 0.78 for the "pre 1978" and "1978-1997" vintages. Since the values in other climate zones are 0.78, this is likely a copy-paste error. This is the same comment as NURS36-Envelope, which implies that the table was copied from that spreadsheet.

LAB28-Envelope (J20)

The SHGC value for the vintage 1998-2007 from Title 2001Table 1-H for a window-to-wall ratio of 31-40% and non-north, is incorrect. It is shown as 0.36 but should be 0.31.

LAB29-Envelope (C3:D12)

It is not clear why a laboratory building would use metal-framed walls and metal decking for a built-up roof. What is the basis for this assumption? Do we have any data to substantiate this assumption?

LAB30-Envelope (H4:X4)

The vintage 1978-1997 wall U-values based on Title 24 1986 were not verified because it was not clear how the values were determined. Please add a description of the process to convert from the values in the table cited to the value shown.

LAB31-Envelope (n/a)

Several envelope table issues were reported for the small hotel that also apply to the hospital prototype table:

- SHSC32 – roof replacement should be weighted
- SHSC37 – use non-north for all windows
- SHSC37 – consider weighting all standards for historic values
- SHSC39 – shouldn't average single and double pane window criteria since non-physical

For brevity, they will not be repeated.

LAB32-HVAC System (G4)

The 2025 Nonresidential and Multifamily Alternative Calculation Method Reference Manual, Table 1, has multiple HVAC system standard designs related to laboratories:

Table 1: Nonresidential HVAC System Map

Space Type	Above-Grade Floors	Climate Zone	System Cooling Capacity	Standard Design
Covered process laboratory with design maximum exhaust 20,000 cfm or less or meets an exception[5] and total building	No limit	All	No limit	System 7b – SZVAVHP

conditioned area less than 25,000 ft^2				
Covered process laboratory with design maximum exhaust 20,000 cfm or less or meets an exception[5] and total building conditioned floor area 25,000 ft^2 and less than 150,000 ft^2	No limit	All	No limit	System 5 – PVAV
Covered process laboratory with design maximum exhaust 20,000 cfm or less or meets exception[5] and total building conditioned floor area ≥ 150,000 ft^2	No limit	All	No limit	System 6 – VAV
Covered process laboratory design maximum exhaust greater than 20,000 cfm and does not meet exception[5] with total building conditioned floor area less than 150,000 ft^2	No limit	All	No limit	System 14a – ACSZVAV
Covered process laboratory design maximum exhaust greater than 20,000 cfm and does not meet an exception[5] with total building conditioned floor area 150,000 ft^2 or greater	No limit	All	No limit	System 14b – WCSZVAV

With the footnote explaining the exception:

5) Exceptions pertaining to covered process laboratory systems include:
i) Existing systems without heating or cooling that are altered,
ii) Systems in climate zone 7, 15
iii) Systems dedicated to vivarium spaces or spaces classified as biosafety level 3 or higher.

Since 14a was selected, the use of over 20,000 cfm and less than 150,000 ft2 makes sense, but the "does not meet the exception[5]" is not explained.

LAB33-HVAC System (G3)

For the existing vintage, the HVAC system does not include any energy recovery, based on the note. That may or may not be valid since many labs that have very high exhaust flows may choose to use energy recovery even if not required by code.

LAB34-HVAC System (D21:D24)

The existing and new total static pressure of 5.58 in w.c. and 5.05 in w.c. need to be better justified than simply citing a PNNL source that does not describe laboratories. This seems too low for a lab.

LAB35-HVAC System (F15)

For new construction, the source should be Title 24-2025, not Title 24-2013.

LAB36-HVAC System (n/a)

The exhaust system in a laboratory is often very complex, with dedicated exhaust fans, manifolded or non-manifolded exhaust systems, heat recovery, and the importance of the exit velocity of the plume. The HVAC system description should describe the exhaust system specifically and how it is being modeled.

LAB37-HVAC System (F21, F25)

A fixed value of 0.9 for the fan motor and drive efficiency is not correct, and instead, the size of the motor and ACM 2025 Table 13 should be used. This is described as the process for existing vintages but not for new construction. The note indicating that this is an average does not explain whether the average value or a calculated value should be used in the prototype models. This is the same comment as LHSC52 and OFFM43.

LAB38-HVAC System (G23:G24)

The new construction fan power per airflow of 1.013 W/cfm for VAV cites "ACM 2025, Table 14 and 15," but no additional explanation is provided. Special filtering may be justified for a lab.

Table 14: Additional System Fan Power Allowance, in W/cfm by System Type

System No.	≤ 5,000 cfm	> 5,000 cfm; ≤ 10,000 cfm	> 10,000 cfm
MZ-VAV (Systems 5 and 6)	0.205	0.174	0.159

System No.	≤ 5,000 cfm	> 5,000 cfm; ≤ 10,000 cfm	> 10,000 cfm
All other (Systems 1, 3, 7, and 9)	0.209	0.182	0.162

Table 15: Adjustment Credits (Multizone VAV) (W/cfm)

Device	≤ 5,000 cfm	> 5,000 cfm; ≤ 10,000 cfm	> 10,000 cfm
Return of exhaust systems required by code to be fully ducted	0.089	0.100	0.116
Exhaust filters, scrubbers, or other exhaust treatment (calculation required, see note)	0.177	0.198	0.231
Particulate filtration credit: MERV 16 or greater and	0.265	0.280	0.333

Device	≤ 5,000 cfm	> 5,000 cfm; ≤ 10,000 cfm	> 10,000 cfm
electronically enhanced filters			
Carbon and other gas-phase air cleaners (calculation required, see note)	0.176	0.188	0.224
Biosafety cabinet (calculation required, see note)	0.177	0.198	0.231
Energy Recovery (included only if standard design requires heat recovery)	0.374	0.318	0.289

This is a similar comment to NURS52.

LAB39-Ventilation (A28:G36)

The exhaust hood table contains calculations, but the results do not seem to be incorporated into the table on outdoor airflow. The calculation of exhaust airflows is often the biggest drivers in laboratory HVAC design and ventilation airflow.

LAB40-Ventilation (C29)

The column heading "Max Exhaust Flow Rate (cfm/ft) [T24, 2025, Table 140.9-C]" references a table that is about kitchen hoods, not laboratory hoods. Also, the value used in the table 250 cfm/ft is actually not in that table, either.

LAB41-Ventilation (C30:C36)

The flow rate through a fume hood is shown as 250 cfm, but it is more likely to be much higher than that. A flow rate of 800 CFM or even 1289 CFM should be used based on the 2019 CASE report on hoods, Table 29 shows:

Hood Characteristics

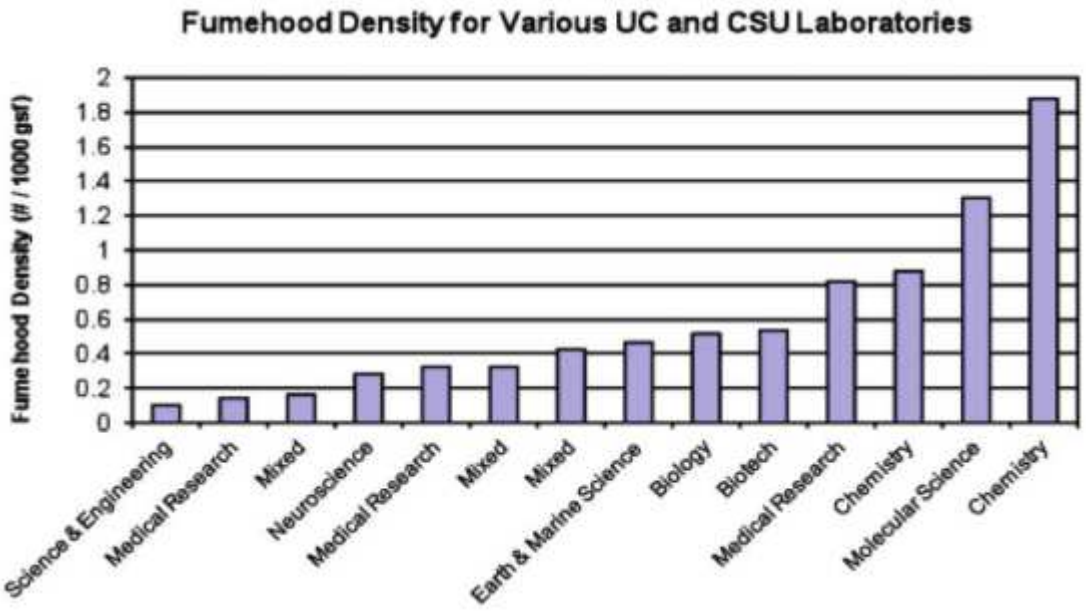
Face Velocity		100	fpm
Hood Length		6	feet
Hood Workspace Depth		24	inches
Hood Workspace Width		5.33	Feet
Closed Height		6	Inches
Sash Stop Height		18	inches
Sash Height		29	inches
Closed Flowrate (Cal/OSHA)		257.8	CFM
Min Hood Flow (Z9.5)		266.7	CFM
Open Hood Flow		800	CFM
Fully Open Flowrate		1,289	cfm
Present Value Lifetime Cost	$	4,140	per hood
Base Occ Sash Position Avg		0.42	
Post Occ Sash Position Avg		0.12	
Base Unocc Sash Position Avg		0.37	
Z9.5 Min	25		CFM/hood area

LAB42-Ventilation (D29:D36, F29:F36)

The spreadsheet cells in D30 to D36 use a simple formula for the hood length of the area of the zone divided by 1000. This is likely based on the note in F30 that says "~1 fume hood per 500–1,500 ft^2 for research or academic lab buildings," but the number of fume hoods and the length are being confused. Typical fume hoods are 6 feet long (they can range from 4 ft to 12 feet), so these values should be multiplied by 6.

LAB43-Ventilation (F29:F36)

The note says "~1 fume hood per 500–1,500 ft^2 for research or academic lab buildings," which is 0.66 or 2.0 hoods per 1000 ft2. PG&E's High Performance Laboratories, Figure 8.01, shows:



This shows a range of fume hood densities from 0.1/1000 gsf to 1.9/1000 gsf, which shows that the type of laboratory is an extremely important metric for the number of fume hoods installed. It is critical for the prototype to describe the type of lab or labs that it is representing due to the very large range of possible fume hood densities. The best solution may be for the prototype to have different lab zones that vary in size according to the proportion of each type of laboratory facility. Then, for each lab zone, have the appropriate number of fume hoods for that type.

LAB44-Ventilation (n/a)

Laboratory exhaust is not only about the fume hoods, but there are other types of exhaust, such as snorkels and containment cabinets, airlocks, as well as general exhaust. These other types of exhaust systems should be counted.

LAB45-Ventilation (G2:G5, G12:G15)

The column is described as "Required exhaust rate-CFM/ft^2- (T24, part 6 2025, Table 120.1-B – minimum exhaust rates)" but this is the MINIMUM exhaust rate of 1.00 CFM/ft2 and is unlikely to be the rate of exhaust in most labs.

LAB46-Ventilation (n/a)

Fume hood sash position and control are extremely important to the amount of exhaust that occurs for each fume hood. The Schedules tab includes three schedules, but these

are not referenced here where they should be referenced. It is not clear how the fume hood sashes should be modeled. The three schedules are called:

- Lab Exhaust CAV
- Lab Exhaust VAV Manual Sash Ctrl
- Lab Exhaust VAV Auto Sash Ctrl

LAB47-Ventilation (n/a)

Overall, the modeling of ventilation for a laboratory is very complicated and should include exhaust from fume hoods, biological safety cabinets, and snorkels; pressurization of zones various zones; ventilation of storage for compressed gas; ventilation requirements for animals used in the laboratory; hazard level; exhaust requirements for special equipment; humidity control for requirements for measurement equipment; airlocks; emergency ventilation requirements; stack height; variable vs. constant volume; and other factors. A bottom-up analysis may be more justifiable for establishing the ventilation requirements.

LAB48-Thermostat (B4:Q4, B6:Q6)

The heating and cooling setpoints and setbacks for the lab and office match the values from Title 24 Appendix 5.4B. But what is not clear is why the lab should not have a constant heating setpoint of 70F and a cooling setpoint of 75F without any setback. This implies either that the lab is never unoccupied (which is not the case according to the lab occupancy schedule) or that the function of the lab does not allow for a setback in heating or cooling. No justification for the lack of a setback in heating or cooling is provided.

LAB49-Thermostat (B3:Q3, B5:Q5)

The setpoints used 70 F/75 F imply that the temperature is standard comfort conditions, which may not be the case depending on the type of lab. Labs that contain animals for testing often need to have setpoints that range from 64 to 85, according to High Performance Laboratories – A Design Guidelines Sourcebook, which is quoting the Institute of Laboratory Animal Resources. That publication also shows that the standard difference between heating and cooling setpoints is 2 F and good practice is 5 F:

Table 8.11:
Benchmarks for laboratory temperature setpoints
(Source: Integra Group, Labs 21)

$$Lab\ Temp\ Setpoint\ Range = Cooling\ SP\ (F) - Heating\ SP\ (F)$$

PG&E Territory		
Standard	Good	Better
2	5	10

That publication also states:

> **Cooling & Heating – Temperature & Humidity Set Points**
> Temperature and humidity set points in laboratory spaces are driven by human comfort and laboratory function (experimentation/equipment requirements). Laboratory users and planners sometimes call for tight tolerances based on laboratory function, without evaluating whether these are actually required. Tight tolerances can increase energy use due to reheat and humidification. It is recommended that tolerances tighter than those required for human comfort (e.g., based on ASHRAE Standard 55), be carefully evaluated and explicitly justified.
>
> At the Global Ecology Center at Stanford, equipment requiring tight tolerances (70F +/- 1F) was grouped into a dedicated area so that other areas of the lab could be controlled to wider tolerances (73F +/- 5F) with some rarely accessed freezers and growth chambers actually relocated to a minimally conditioned adjacent structure controlled to 55F–95F.

Another source, the Laboratories for the 21st Century: Energy Analysis used 72 F/74 F.

LAB50-Thermostat (n/a)

The lack of humidity control setpoints implies that no humidity control is being done. For many types of labs, that would not be acceptable. For many types of lab instruments, humidity control is required, and depending on the lab activities, humidity may need to be controlled. In addition, with the large amount of outside air that is necessary in many labs, this would result in indoor humidity issues in climates with humid periods.

LAB51-Thermostat (n/a)

Some spaces in laboratory buildings may have special requirements for temperature control. For some biological labs, it might make sense to keep the temperature cool to prolong the time an unrefrigerated sample is viable. Some types of physical science laboratory equipment may need to be kept in a cooler than usual environment. Animal labs may need to keep the temperatures closer to the natural environment of the animals, which could be cooler or warmer than for human comfort. Some storage rooms, stairways,

and mechanical/electrical rooms may not be cooled at all.

LAB52-Water Heater (C10:D24)

These cells have links to an external copy of a similar (probably earlier version) of the spreadsheet. For example, cell C10 shows:

> ='N:\Prj_SUS\2023\S23668 SCE CA Prototypes Development\Technical\NR\Laboratory\[Lab_Prototype_Scorecard_REV4.xlsx]Zones'!F3

It should simply link to

> =Zones!F3

LAB53-Water Heater (E10:E12, E17:E20)

While the hot water usage of 0.18 gal/hr per person for laboratories and offices is consistent with the cited Appendix 5.4A value, it seems likely that additional hot water may be required for certain types of laboratories that require washing of glassware and equipment or for use in hot-water baths which are commonly used in biological and chemical laboratories to keep samples at a constant elevated temperature.

LAB54-Water Heater (B10:B24)

The use of both electric and gas water heating equipment for new construction in the same building is quite unusual, except in the case of small electric water heaters being used under remote washrooms for sinks. This is unlikely in a laboratory that already has extensive piping to serve all labs. Based on this, all water heating should probably be gas. This is the same comment as LHSC61-Water Heater and SHSC59-Water Heater.

LAB55-Water Heater (D10:D12, D17:D20)

The number of people per space in the cited Appendix 5.4A is 10 people/1000 sqft instead of the 25 people/1000 sqft used. Since the people density is part of the estimate in the spreadsheet, it is not clear if the 0.18 gal/hr per person value used is appropriate.

LAB56-Water Heater (n/a)

For some laboratory applications, higher temperatures may be required and would likely be served by boilers.

LAB57-Water Heater (B36)

The cold-water supply temperature of 55°F shown in the scorecard is consistent with the citation of the ACM 5.9.2 Rated Capacity, but better data is available that may be closer to how water heaters are actually sized using the local water mains temperature. The Appendix 5.4B (called CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx) includes a WaterMain tab and WaterMainMonth tab that show climate-specific values for the entering water temperature. The lowest of these values for each climate zone is more likely to be used for sizing water heaters; the monthly values vary from 41.4 in CZ16 to 63.2 in CZ15. Using these values would have a large impact on the sizing of the water heater. This is the same comment as LHSC64-Water Heater, SHSC61-Water Heater, and HOSP39-Water Heater.

LAB58-Water Heater (C3)

The efficiency of the gas water heaters for both existing and new construction may not be the best assumption. Older water heaters may have been less efficient initially or may decrease in efficiency due to corrosion of the heat exchanger or burner.

LAB59-Interior Lights (C4:C5, F4:F5)

While the lighting power densities for new construction and existing are consistent with the referenced versions of Title 24, the additional lighting power cells are shown as empty. The empty cells imply that these values are zero, but that is very unlikely for laboratory spaces and somewhat unlikely for office spaces.

For new construction, the table from Title 24 shows:

Building Type/Use	Primary Function Area	Allowed Lighting Power Density for General Lighting (W/ft²)	Additional Lighting Power Qualified Lighting Systems	Additional Lighting Power Additional Allowance [3] (W/ft², unless noted otherwise)
Laboratory, Scientific	NA	0.90	Specialized Task Work[8]	0.35
Office Area	> 250 square feet	0.60	Decorative/Display and Portable lighting for office areas[6]	0.20

With referenced footnotes

6. Portable lighting in office areas includes under shelf or furniture-mounted supplemental task lighting qualifies when controlled by a time clock or an occupancy sensor.

8. Specialized task work – Lighting provides for small-scale, cognitive or fast performance visual tasks; lighting required for operating specialized equipment associated with pharmaceutical/laboratorial activities.

It seems like the 0.35 W/ft2 for laboratories should be added, given the specific reference to laboratory activities in the footnote.

The additional power for office areas is also reasonable for a model, at least for part of the offices.

For the existing buildings, similar tables show:

Footnote number	Type of lighting system allowed	Allowed lighting power density. (W/ft^2 of task area unless otherwise noted)
1	Specialized task work	0.20 W/ft^2

LAB60-Interior Lights (G4:G5)

No justification is used for using T24-2016 for existing lighting power densities. It seems a bit optimistic for many of the space types in a laboratory to have their lighting systems replaced that quickly. Given the high cost of operating a lab and that that is almost all based on the internal equipment and HVAC system, it is likely that interior lighting retrofits would be far down on the list of improvements. This is very similar to HOSP42-Interior Lights.

LAB61-Interior Lights (C3, F3)

The heading is labeled

"Additional LPD for PorTable Lighting (W/ft^2)"

The use of capitalized T in the middle of "portable" seems odd. Also, I'm not sure why it is even called this, since the portable lighting is generally just referenced for offices, and the more general term is "Additional Lighting Power Allowance."

LAB62- Interior Lights (G4:G5)

Using Title-24 2016 for all existing buildings may not be as accurate as taking an average of multiple versions of Title 24 covering recent vintages, assuming that older buildings have

replaced their lighting in the last 30 years (or some other time period). This is a similar comment to OFFM55-Interior Lights.

LAB63-Ext Lights (B10:C10)

The parking area reported 85,813 ft2 is based on the formula "=B21/250*400," but this implies that a 400 ft2 parking place is present for each 250 ft2 of building area, whereas the note in cell B10 says it should be for each 830 ft2. Either the note or the formula is wrong. Both the 250 and 830 appear on the 2019 Outdoor Lighting CASE report:

Table 38: Assumptions for Converting Indoor New Construction Forecasts to Outdoor Spaces for General Hardscape – Table 140.7-A

Assumptions for Statewide Estimates - General Hardscape		
General Hardscape	Assumptions	Area Multipliers for Construction S.F.
for Large Office, Small Office, Food, Restaurant, College	1 parking space per 250 sf of gross building area	1
for Hotel, Retail, School, Other	1 parking space per 360 sf of gross building area	0.7
for NR Warehouse, Ref. Warehouse	1 parking space per 830 sf of gross building area	0.3

But neither corresponds to a laboratory.

Many codes have requirements for this, such as:

https://www.codepublishing.com/CA/SanCarlos/html/SanCarlos18/SanCarlos1820.html

which shows a parking spot for each 800 ft of laboratory area.

LAB64-Ext Lights (B11:C11)

The lighting per building entrance is based on two entrances on cell C11, but cell B11 says that the model has 14 entrances. Since the model was apparently for an office building rather than a laboratory, it would probably make sense to find another source for this assumption. Two entrances seem like too few for this size building.

LAB65-Ext Lights (B12:C12)

The hardscape ornamental lighting of 0.1 ft2 per 1 ft of gross building area is accurate, but is only applied to a fraction of the building in the CASE report referenced, as seen by the table below:

Table 36: Assumptions for Converting Indoor New Construction Forecasts to Outdoor Spaces for Specific Applications – Table 140.7-B

Assumptions for Statewide Estimates - Specific Applications											
		Applied to % of Building S.F. in Category									
Lighting Allowance	Assumptions	Office, LG & SM	Retail	Restaurant	Food (Grocery)	Warehouse, Ref & NR	Hotel	School	College	Other
Hardscape Ornamental Lighting	**0.1 SF per SF of gross building area**	50%	50%	50%	25%		50%	25%	25%	5%

Given that the values of 50% are shown for buildings that would want to impress visitors, it is more likely that either 0%, like a warehouse, or 5%, for others, should be chosen. If 5% is used, then the exterior lighting power should be 2 W instead of 38 W.

LAB66-Ext Lights (G10)

The formula in cell G10 has a fixed value for the area of 65520 instead of using the area in cell C10.

LAB67-Ext Lights (B13:C13)

The building façade lighting assumptions in B13 imply the building height is 28 ft, but it is 39 ft, so 30 ft should be used since it is the maximum height lit. The 30 ft value is actually used in the formula in C13, so the note needs to be fixed. But also a multiplier of 0.5 is used for a factor of only half the perimeter is lit, but based on the CASE report:

Table 36: Assumptions for Converting Indoor New Construction Forecasts to Outdoor Spaces for Specific Applications – Table 140.7-B

Assumptions for Statewide Estimates - Specific Applications											
		Applied to % of Building S.F. in Category									
Lighting Allowance	Assumptions	Office, LG & SM	Retail	Restaurant	Food (Grocery)	Warehouse, Ref & NR	Hotel	School	College	Other

Building Facades	**30' building height, 2 floors per building** (20% of applicable facades are lit)	25%	50%	50%	25%		50%	25%	25%	5%	

Given that the values of 50% are shown for buildings that would want to impress visitors, it is more likely that either 0%, like a warehouse, or 5%, for others, should be chosen. If 5% is used, then the exterior lighting power should be 81.9 W instead of 819 W.

LAB68-Ext Lights (A14:C14)

The assumption of 1788 ft2 of non-sales canopy area is not consistent with the shading geometry description on the Prototype tab that indicates no shades.

LAB69-Ext Lights (A8)

No justification is provided for the use of Lighting Zone 2 (LZ2).

LAB70-Ext Lights (D3, B8)

The reference in these two cells should be updated. The reference to Title 24-2022 can be removed since Title 24-2025 is being used. The reference to the PNNL 90.1-2010 TSD should include the actual report number PNL-20405.

LAB71-Ext Lights (n/a)

Where are the exterior lighting assumptions for the vintage models? One assumes that those should be higher. This is the same comment as LHSC72, SHSC67, and HOSP44.

LAB72-Equipment (E3:E4)

The reference to T24 2022 Appendix 5.4A should be replaced with the latest version, which is from 2025 and contains the same values.

LAB73-Equipment (D3)

The refrigeration density of 0.28 W/ft2 is not consistent with T24 2022 Appendix 5.4A (or 2025), which shows zero. The note in F3 is confusing as to what comes from where "Appendix 5.4A: Gas equipment=13 Btu/h-ft^2--CEC model (Lab) has the Gas equipment for

51 Btu/h-ft^2 and Refrig load for 0.28 w/ft^2---CPUC supports this data”

LAB74-Equipment (B4)

The 0.7 W/ft for office plug loads is not consistent with the source "T24 2022 - Appendix 5.4A" but is shown in the reference "Plug load and schedules project with CEC, 2025." This reference should also be added to the Notes tab. The reference should be cited as:

Proposed Updates to Equipment Power Densities and Schedules in the 2028 Energy Code Nonresidential and Multifamily Alternative Calculation Method (NRMFACM) Reference Manual. Contoyannis, et al. August 7, 2025. TN265693.

LAB75-Equipment (I11:I12, D19)

References to PNNL ASHRAE 90.1-2013 or 2022 laboratory prototype, or to "same as PNNL assumptions," imply that PNNL had a laboratory prototype, but that is not the case. The PNNL prototype that has some historical connection is for the medium office building. These references overstate the connection to PNNL.

LAB76-Equipment (B11:B12, A15:D19)

Assuming three elevators in this size building is reasonable, but the justification provided is not for a laboratory building. Please justify the use of three elevators.

LAB77-Equipment (B11:B12, A15:D19)

I would expect one of the three elevators to be a larger freight elevator capable of moving larger laboratory equipment to upper floors. This would impact the elevator load, lighting and ventilation.

LAB78-Equipment (H11:H12, A24:E29)

The design load for elevators is shown as the sum of the lift, light, and fan. The formula in the table ignores the lighting power in this sum. The elevator lift is shown but not justified for any of the three vintages. The elevator fan value is not justified and does not seem to be based on the fan power limit shown in cells F11 and F12. The lighting power shown in the

table does not seem to be consistent with the powers in the C11 and C12. Also, if the ACM says to use 10 kW.

LAB79-Equipment (D18)

The lighting shutoff for elevators should be after 15 minutes according to 120.6 (f)(3):

3. When the elevator cab is stopped and unoccupied with doors closed for over 15 minutes, the cab interior lighting and ventilation fans shall be switched off until elevator cab operation resumes.

Not 5 minutes as shown. This is the same comment as SHSC73.

LAB80-Equipment (n/a)

Given the amount of computation done to analyze laboratory results, it is very likely that a small computer room with high internal loads would be present in this type of building.

LAB81-Equipment (n/a)

The document "Market Assessment of Energy Efficiency Opportunities in Laboratories" includes a very detailed assessment of plug loads in laboratories. It includes a long list of equipment that is commonly used in laboratories:

TABLE 1: EQUIPMENT INVENTORY LIST

CATEGORY	TYPE
Refrigeration	-80°C Freezer -20°C Freezer Refrigerator
Fume Hoods	All Fume Hoods Variable Air Volume
Microscopes	Fluorescence Microscopes Confocal Microscopes Electron Microscopes
Benchtop Equipment	Heating Block Water Bath Centrifuge PCR Machine Magnetic Stir Plate Vacuum Pump Water Distiller
Large Laboratory Equipment	Shaker Table Autoclave Gas Laser qPCR Machine NMR Mass Spectrometer Gas Chromatograph HPLC FACS Incubator Tissue Culture Hood Sonicator Vacuum Chamber Air Table
Hospital Equipment	MRI CT Scanner X-Ray Machine
Environmental Room	Warm Room Cold Room

The report surveyed many different types of laboratories and asked for the number of each type of equipment in each type of lab.

In addition, it provided different utilities that were found in labs:

TABLE 72: UTILITIES FOUND IN LABORATORY BUILDINGS

	PERCENT AFFIRMATIVE RESPONSES
Compressed Air	97.5%
Distilled Water	93.8%
Gas Storage	92.0%
Vacuum System	93.2%
Supplied Gas	94.7%
Process Chilled Water	87.7%
Supplied Ultra-Pure Water	83.3%
Dry Solvent Delivery System	37.5%
Point Source Exhaust System	83.3%

The following table could be used to adjust schedules.

TABLE 85: ESTIMATED ENERGY CONSUMPTION RANGES FOR LABORATORY EQUIPMENT

EQUIPMENT TYPE	AVG HRS OF USE PER DAY	ENERGY CONSUMPTION RANGE (PUBLISHED)	ENERGY CONSUMPTION RANGE (KWH/YR)
-80°C Freezer	24	3900 – 13,666 kWh/year	3900 – 11,100
-20°C Freezer	24	1690 – 4876 kWh/year	1690 – 4876
Refrigerator	24	199 - 2686 kWh/year	199 – 2686
Fume Hood	24	30 - 60 kWh/day	10,950 – 21,900
Fluo Microscope	3	0.5 – 1 kWh/day	183 – 365
Centrifuge	3	3.2 – 57 kWh/week	166 – 2964
Water Bath	13.5	2025 – 3850 kWh/year	2205 – 3850
Heating Block	3	243 kWh/year	243
PCR Machine	4	788 kWh/year	788
Incubator	24	13.1 – 167 kWh/week	681 – 8684
Shaker	3	42 kWh/week	2184
Autoclave	3	32 – 630 kWh/week	1664 – 32760
Vacuum Pump	3	0.09 kWh – 7.5 kWh/day	33 – 2730
Tissue Culture Hood	4	60 – 88 kWh/week	3120 – 6862

In combination with the following counts, end-use estimates could be made:

TABLE 74: ESTIMATED NUMBER OF LABORATORIES (ADJUSTED) BY MARKET SEGMENT IN CALIFORNIA

MARKET SEGMENT	ESTIMATED AVERAGE SQ FT (M) PER LAB	ESTIMATED NUMBER OF LABS
Academia	4,000	6,200
LSR	5,800	11,700
Hospitals	4,600	1,700
Other	5,700	540
Total		**20,100**

Or using the following table:

TABLE 75: ESTIMATED TOTAL NUMBERS OF LABORATORY EQUIPMENT IN CALIFORNIA

EQUIPMENT	AVERAGE NUMBER PER LAB REPORTED IN CA	ESTIMATED TOTAL NUMBER IN CA
-80°C Freezer	2.9	58,000
-20°C Freezer	3.7	74,000
Refrigerator	4.7	95,000
Fume Hood	3.0	60,000
Fluorescence Microscope	1.7	34,000
Heating Block	3.0	60,000
Water Bath	2.6	52,000
Centrifuge	3.8	76,000
PCR Machine	2.2	44,000
Magnetic Stirrer	3.0	60,000
Vacuum Pump	2.1	42,000
Shaker Table	1.2	24,000
Autoclave	0.8	16,000
Incubator	3.0	60,000
Tissue Culture Hood	1.7	34,000

Using this bottom-up approach from data from this survey might provide a more accurate value for laboratory plug loads.

LAB82-Schedules (AB37)

The schedule values were compared to those that came from Title 24-2025 Appendix 5.4B, and they all matched except hour 24 for office occupancy, which shows zero but should be 0.05.

LAB83-Schedules (C4:C45)

The description of the zones that apply to schedules is inconsistent with the description on the Zones tab.

- "Level 2 and 3 Lab Zones" should be "Level 1 through 3 Lab Zones"
- "Level 1- Office Zones" should be "Level 1 and 3 - Office Zones"

LAB84-Schedules (B4:B45)

The source of the schedules should be changed from "T24 2022 - Appendix 5.4B" to the latest version of the appendix spreadsheet "T24 2025 - Appendix 5.4B".

LAB85-Schedules (A28:A36)

Three different exhaust fan schedules are described and are present in the source T24 2025 - Appendix 5.4B:

- Lab Exhaust CAV
- Lab Exhaust VAV Manual Sash Ctrl
- Lab Exhaust VAV Auto Sash Ctrl

It is great to see this level of detail, but no indication of how this is applied or any description of exhaust to this level of detail that these schedules should be applied. This level of detail is important and should be used, but the exhaust hood descriptions on the Ventilation tab do not reference these schedules.

LAB86-Schedules (n/a)

Some labs that support medical testing often have longer hours since hospitals and medical clinics may need them. For this type of lab, I would expect laboratory occupancy and all related schedules to have more than 5% to 10% during nighttime hours.

LAB87-Schedules (E13:AB18)

The lab service hot water schedule and lab elevator schedule show as always zero. This clearly does not make sense since both functions would occur in a laboratory building during occupied hours.